

Relation between physical security and cyber security

Introduction: (Definitions)

Cyber Security:

Cyber Security: Cybersecurity protects digital assets, networks, and systems from unauthorised access, data breaches, and cyber threats. It involves implementing encryption, firewalls, intrusion detection systems, and security protocols to eliminate the risks associated with internet-connected systems and activities.

Physical Security:

Physical Security: Physical security, on the other hand, is the protection of tangible assets, premises, and personnel from physical threats like theft, vandalism, aggression, and any form of unauthorised access. This includes electronic access control systems, surveillance cameras, security guards, lighting, and perimeter fences and barriers.

Key points:

Cyber Security:

- 1. Firewalls**
- 2. Antivirus and Antimalware Software**
- 3. Intrusion Detection and Prevention Systems (IDPSs)**
- 4. Secure Sockets Layer/Transport Layer Security (SSL/TLS)**
- 5. Two-Factor Authentication (2FA)**
- 6. Security Awareness Training**
- 7. Regular Updates and Patch Management**

Physical Security

- 1. Access Control Systems**
- 2. Perimeter Security**
- 3. Surveillance Systems**
- 4. Intrusion Detection Systems**
- 5. Security Personnel**
- 6. Security Alarms**

Current technology outlook:

- Cybersecurity attracts an enormous amount of attention due to cyberattacks that are publicized daily. As more devices are connected to the Internet, they become attractive targets for criminals; therefore, the attack surface increases exponentially.
- As with any technological advancement, connected devices have become a common target for cyber criminals hoping to steal valuable data or even cause potential destruction of property. Connectivity comes at a cost, and protecting connected devices through effective cybersecurity now goes hand-in-hand with physical security, creating a convergence of the physical and digital domains that reflects the increasingly interconnected state of the world in which we function.

The relation:

- Physical security acts as the first point of engagement by limiting who has access to the areas where data is stored. This includes the setup and maintenance of security cameras and round-the-clock physical protection from security guards. Physical security is also personal security. Defending and being situationally aware of the activities of your executive team is essential to protecting them and their families.
- When it comes to the cyber world, using the information that can be found on both the surface and the dark web, in addition to information exchanged within closed forums and social channels, can help bolster defenses. It's not just about biometrics, passwords, and secure WiFi. Physical security, executive protection, and cybersecurity are more effective when they share relevant, timely, and actionable information from multiple sources.
- **Symbiotic relationship between cyber security and physical security**

Advantages:

- Address a diverse range of threats and ensure that no vulnerability is left unguarded.
- Set up a resilient and adaptable defence against potential risks.
- Through regular updates, training programs and security audits, you ensure that your organisation can effectively mitigate new and emerging threats.

Challenges:

- **Integration Complexity:** Combining physical and cyber security systems requires seamless integration of hardware, software, and protocols, which is technically and logistically complex.
- **Insider Threats:** Insiders with legitimate access to physical or digital assets can exploit their knowledge and privileges to bypass security measures.
- **Increased Attack Surface:** The convergence of physical and cyber systems, especially through IoT devices, expands the attack surface for adversaries.
- **Resource and Budget Constraints:** Balancing investments in both physical and cyber security often strains budgets, especially for organizations with limited resources.
- **Maintenance and Reliability:** Ensuring the continuous operation of integrated physical and cyber security systems requires ongoing maintenance and updates.

Conclusion:

- Even though physical and cybersecurity are inherently connected, many organizations still treat these security functions as separate systems.
- Now the problem comes down to governance, making it a priority to create a single body for security policies and bringing physical security and cybersecurity teams together to build strength in your organization.
- An integrated security architecture offers a foundation for connecting the physical and cyber worlds through intelligence sharing, visibility, control, and automation.
- As we use more technology in our daily lives, the more there is a need for CPS to help protect your organization from accidental and potentially malicious.